

22CH203 – UNIT 1- ELECTROCHEMISTRY

LECTURE MATERIAL – TOPIC - 5

Topic: Ion selective electrodes - construction and working of glass electrode

Ion-selective Electrodes

The electrodes which have the ability to respond to certain specific ions and develop a potential with respect to that species only in a mixture are called ‘Ion Selective Electrodes’.

These electrodes use a membrane which is sensitive to particular chemical species. These are also referred to as ‘membrane electrodes’.

Eg : Glass membrane is H^+ ion selective only and does not respond to any other ion.

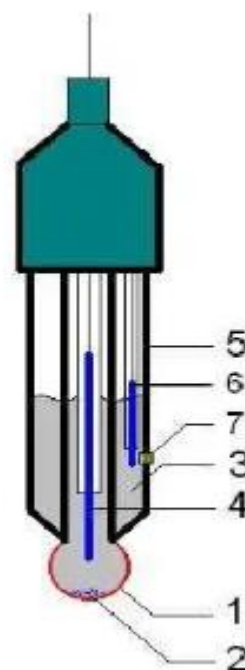
Glass Electrode

It is the most versatile electrode, highly sensitive to hydrogen ion concentration. The potential of this electrode depends linearly on the pH of the solution. It is easier to handle glass electrode than the hydrogen electrode and is widely used in pH measurements. The glass electrode is fabricated by dipping a silver wire plated with silver chloride into a 0.1 M HCl solution contained in a glass tube with a thin glass membrane or glass bulb at its end.

Construction:

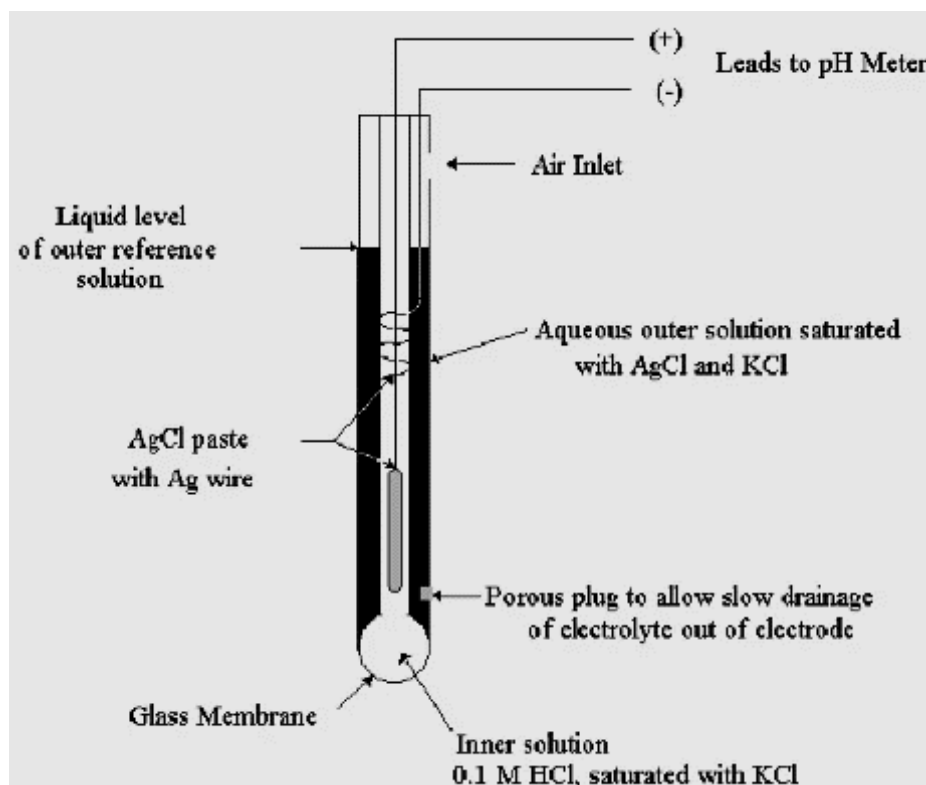
Scheme of pH glass electrode

1. a sensing part of electrode, a bulb made from a specific glass, relatively lowmelting point and high electrical conductivity.
2. sometimes the electrode contains a small amount of AgCl precipitate inside the glass electrode.
3. Internal solution, usually 0.1M HCl for pH electrodes.
4. Internal electrode, usually silver chloride electrode or calomel electrode



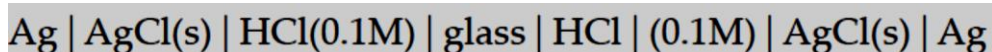
5. Body of electrode, made from non-conductive glass or plastics.
6. Reference electrode, usually silver chloride electrode or calomel electrode
7. Junction with studied solution, usually made from ceramics or capillary with asbestos or quartz fiber.

Working:

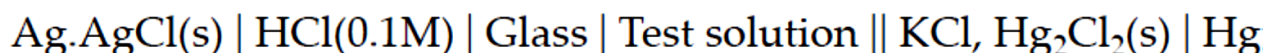


Glass Electrode

1. It consists of thin walled glass bulb containing AgCl-Coated Ag electrode or simply at platinum electrode in 0.1M HCl.
2. HCl in the bulb furnishes a constant H^+ ion concentration. Thus it is a silver – silver chloride electrode, reversible with respect of chloride ions.
3. When two solutions of different pH values are separated by a thin glass membrane, there develops a difference of potential between two surfaces of the membrane. The potential difference developed is proportional to the difference in pH value.
4. The glass membrane functions as an ion – exchange resin, and an equilibrium is set up between the Na^+ ions of glass and H^+ ion concentration.
5. The potential difference varies with the H^+ ion concentration
6. It is represented as follows,



In order to measure the hydrogen ion concentration of the test solution, the glass electrode is coupled with a reference electrode such as a calomel electrode and the cell will be represented as



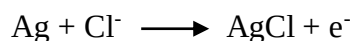
The advantage of using the glass electrode is that it can be used in the presence of strong oxidizing and reducing agents, in viscous media and in substances like proteins which seriously interface with other types of electrodes.

Determination of pH of a solution by coupling glass electrode with saturated calomel electrode

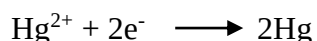
In a container, HCl solution of unknown concentration should be taken. On left hand side, glass electrode containing 0.1M HCl solution should be dipped into HCl solution of unknown concentration. On right hand side, saturated calomel electrode should be set up. Now both the electrodes are coupled through potentiometer.

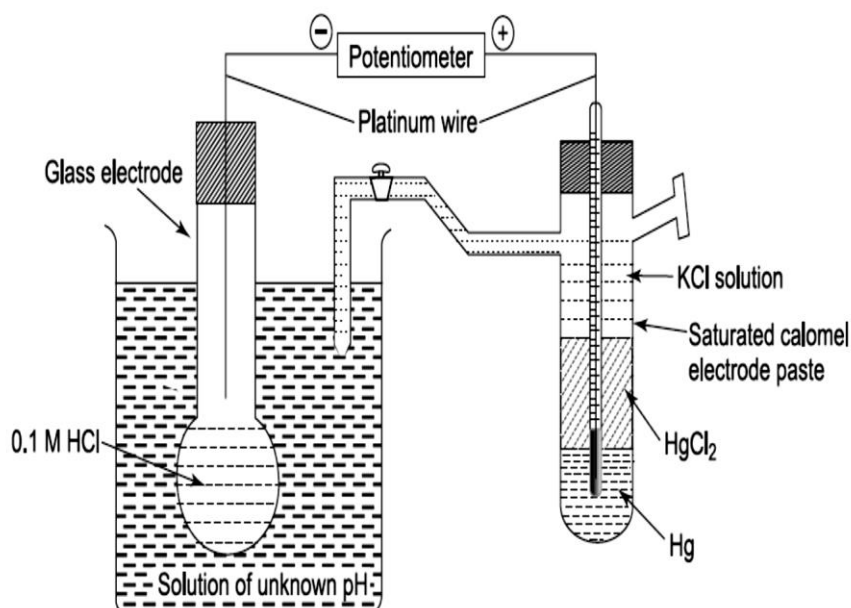
It is important to note that, between the outer and inner surfaces of the glass bulb, of the glass electrode, a potential difference is generated due to the difference in the pH values of two different HCl solutions. Greater the difference in pH, higher will be the potential difference. This potential difference is the electrode potential of the glass electrode. The potential difference between anode and cathode is measured with potentiometer.

At anode:



At cathode:





The cell can be represented as follows:



The emf of the cell is measured potentiometrically at 25°C. From the emf, the pH of the unknown solution is calculated as follows

We know that,

$$E_{\text{cell}} = E_{\text{right}} - E_{\text{left}}$$

$$E_{\text{cell}} = E_{\text{std. calomel electrode}} - E_{\text{Glass electrode}} \longrightarrow \text{Eq. 1}$$

At 25°C, the potential of the glass electrode can be calculated as

$$E_G = E_G^0 + \frac{2.303RT}{F} \text{ pH}$$

$$E_G = E_G^0 + 0.0591 \text{ pH} \longrightarrow \text{Eq. 2}$$

By substituting the value of E_G in the Equation 1, pH can be calculated.

$$E_{\text{cell}} = 0.2422 - (E_G^0 + 0.0591 \text{ pH})$$

$$\text{pH} = \frac{0.2422 - E_G^0 - E_{\text{cell}}}{0.0591}$$



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E_G^0 can be determined by using a solution of known H^+ ion concentration and then by measuring EMF of that cell (Eq 2).

Advantages of glass electrode

- Glass electrode is not affected by oxidizing and reducing agents.
- It can be used in the presence of metal ions, poisons, coloured solutions etc.
- A small quantity of solution is sufficient for the determination of pH.
- The exchange reaction between the Na^+ and H^+ ions is attained immediately.
- It is extensively used in chemical, industrial, agricultural and biological laboratories.
- Electrode of special glass can be used for measuring pH values up to 12.

Limitations

- The glass membrane has an extremely high resistance. Hence, special electronic potentiometers are to be used.
- It cannot be used in strongly alkaline solution ($pH > 10$). In such cases, special type of glass must be used.

Note

- The Na^+ ions on both sides of the structure of the glass membrane are exchanged with those of H^+ ions of HCl kept on the both sides of the glass membrane of the glass bulb.
- The evaluation of E_G^0 becomes necessary because the magnitude of it differs with different varieties of glass electrode for the same concentration of H^+ ions of the experimental solution.
- When an experiment with a glass electrode is completed its inner and outer membrane are depreciated. Hence, the evaluation of E_G^0 for the same glass electrode is necessary every time it is used.

Reference:

Chemistry for Engineering students by Lawrence S Brown and Thomas A Holme (Fourth Edition) Chapter 13